

2 High-brightness LEDs

Currently, high-brightness light-emitting diodes (LEDs) are the most promising light source, as these offer better efficiency, longer life and higher mechanical strength. Because high-brightness LEDs have a typical heat flux of over $100\text{W}/\text{cm}^2$, conventional packaging is not suitable. High junction temperature in LEDs substantially degrades the efficiency, colour quality, reliability and life of the solidstate lighting devices. Failure analysis always shows that the failure is not on the LED itself but on different package components due to high operating temperatures.

Nanomaterials can play a big role in heat dissipation in LEDs. As shown in Fig. 3, a die-attach adhesive not only attaches LED die to the substrate but also provides thermal and electrical conduction between the LED die and the package. Hence the heat-conducting ability of die-attach adhesive is very critical to the performance.

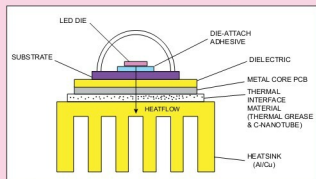


Fig. 3: Improved LED

The existing epoxy adhesive has very low shelf temperature at -40 to -20°C , which is needed to lower the reactivity. This makes it very inconvenient and energy consuming to transport and store the material. Second, most of the epoxy resins need to be cured at a high temperature of above 130°C for a long time, which may damage the LED. The high curing temperature wastes energy and reduces mass production efficiency.

Research work in this area has led to the development of a die-attach adhesive with high thermal conductivity of about $25\text{W}/\text{MK}$, low curing temperature of about 85°C and high shelf temperature of -10°C using nanomaterials and processing technology. This was made possible by using metal nanowires as filler additives in the adhesive matrix. These filler additives can fill the gaps between the original commercial fillers to form a continuous and multichannel heat transmission pathway to enhance the thermal conductivity as

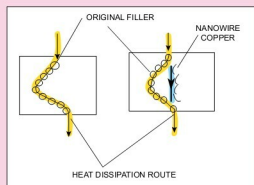


Fig. 4: Die-attach adhesive matrix

shown in Fig. 4.

A series of low-cost environment-friendly (heavy-metal-free) luminescent quantum dots as the down-converting phosphors for high-performance LED devices has been developed. These luminescent, spectrally tunable quantum dots are coupled to a single-crystal LED chip to serve as a colour converter.

Quantum dots are quantised electronic structures in which electrons are confined with respect to motion in all three dimensions. Their properties resemble those of atoms in an electromagnetic cage, rendering possible fascinating novel devices. Typically, the size of a quantum dot is 10 nm and it may contain ten or more atoms. The size generally varies from 4 to 20 nm at room temperature depending on the type of the material. Such drastic reduction in size leads to carrier localisation in all three dimensions.

Using proper combination of quantum dots with different sizes and compositions and with improved package design, it is possible to manufacture high-quality, non-toxic white solidstate lighting devices with high colour rendering index ($\text{CRI} > 85$) and high efficacy ($> 60\text{ lm}/\text{W}$). Quantum-dot technology can significantly save energy and improve the colour quality of conventional LEDs.

There are two main classes of white LEDs: multi- and single-chip devices. Multichip white LEDs (RGB LEDs) constituted by red, green and blue emitting chips show three emission bands and possess a good CRI, good efficiency and well tunable colour. However, the efficiencies of red, green and blue LEDs change over time at different rates. Hence after a high-quality white light is produced initially, the quality of the white light degrades over time. Also, the cost of multichip white LEDs is high.

In comparison, single-chip white LEDs are low-cost and offer a high luminous efficiency, making them suitable for general-purpose lighting in the future. These LEDs are also called down-conversion LEDs, in which the blue light is downconverted into the light of a longer wavelength and the combination of blue and yellow lights is interpreted as white light by the human eye.

However, commercially available white LEDs emit a harsh and bluish cold-white light with poor colour rendering properties. This limits their wide-scale use in indoor illumination. When these white LEDs are used as the backlight of an LCD, the pale white light from blue and yellow hues cannot express the natural colours of objects faithfully in general circumstances.

Quantum dots can be easily optimised in size to a white light spectrum. The small size of quantum dots, which is much smaller than the wavelength of visible light, can also eliminate all light scattering and the associated optical losses. Thus highly luminescent and spectrally tunable quantum dots can be coupled to bright and efficient single-crystal LEDs. This approach utilises quantum dots

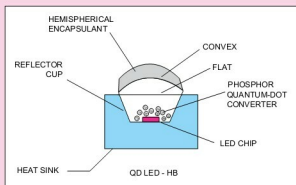


Fig. 5: Quantum-dot high-brightness LED